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Markets in a Spin
A Regime-Aware Regularised Spin-Interaction Model for
Cross-Sectional Equity Returns

Group 7

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Contents

1	Introduction	3
2	Background and Model Foundations	4
2.1	The Ising Model in Statistical Mechanics	4
2.2	From Magnets to Markets	4
3	The Static Ising Model	5
3.1	Binary Market Representation	5
3.2	Limitations	5
4	The Dynamic Ising Idea	5
4.1	Motivation for Time-Varying Interactions	5
4.2	Why Direct Time-Series Modelling Is Difficult	6
5	The Conditional Ising Model	6
5.1	Lagged Conditional Dependence	6
5.2	Logistic Conditional Structure	6
5.3	Rolling-Window Estimation	7
5.4	ℓ_1 -Regularised Estimation and Coordinate Descent	7
5.5	Prediction and Empirical Performance	8
6	Why Binary Spins Are Not Enough	9
7	The Blume–Capel Model	10
7.1	Tri-State Representation	10
7.2	Hamiltonian and Physical Interpretation	10
8	PCA and Latent Spin Encoding	11
8.1	Training Window and SVD	11
8.2	Quantile and Z-Score Thresholding	11
9	Inference and Learning	12
9.1	Intractability of Exact Inference	12
9.2	Mean-Field Approximation	12
9.3	Parameter Identifiability and the d -Centering Constraint	14
9.4	Pseudo-Likelihood Estimation	14
9.5	Mean-Field Log-Partition Approximation	15
10	Regime-Switching Extension	15
10.1	Hidden Regimes and HMM Formulation	15
10.2	Log-Space Forward Filtering	16
10.3	Multi-Step Regime Prediction	16
10.4	Modified EM-Style Learning	17

11 Online Learning and Trading System	18
11.1 Weighted Pseudo-Likelihood Objective	18
11.2 Composite Objective and the Need for Proximal Methods	18
11.3 Transition Matrix Update	21
11.4 Signal Generation	22
11.5 Portfolio Construction and Risk Management	23
12 Walk-Forward Validation	24
12.1 Rolling Window Design	24
12.2 The Three Control Parameters	24
12.3 Fixed versus Dynamic Components	24
13 Empirical Results	25
13.1 Performance Metrics	25
13.2 Interpretation	26
13.3 The Nifty 100 Universe	27
13.4 The Nifty Pharma Sector	29
14 Conclusions	30
A Maximum-Entropy Derivation of the Hamiltonian	31
B Consistency of Pseudo-Likelihood Estimation	32
C Convergence of the Mean-Field Fixed-Point Iteration	32
Code Availability	34

Abstract

Equity returns across stocks are rarely clean or stable: they are noisy, shift with market conditions, and often behave differently from one regime to the next. This report builds a sequence of statistical-physics-inspired models to capture those patterns in the Indian equity market, moving from simple shared structure to discrete collective behaviour and slow regime changes. We begin with a static Ising baseline, explore the conceptual and computational barriers to fully dynamic interaction matrices, introduce a tractable lagged conditional Ising model estimated via rolling-window ℓ_1 -regularised logistic regression, and finally arrive at a regime-aware Blume–Capel system with tri-state latent spins. The final model compresses returns through adaptive PCA, discretises latent scores into $\{-1, 0, +1\}$ spins, learns regime-specific interaction patterns via mean-field pseudo-likelihood, filters latent market regimes with a log-space hidden Markov model, and updates parameters online by mirror descent. Forecasts are mapped into a risk-managed portfolio with explicit volatility control and transaction-cost accounting. Using a true walk-forward protocol on Nifty 50 constituents from November 2017 to January 2025, the Blume–Capel strategy achieves an annualised Sharpe ratio of 1.025 and a maximum drawdown of -25.62% , substantially improving upon both the rolling conditional Ising model and an equal-weight buy-and-hold benchmark.

1 Introduction

Financial markets are systems of many interacting parts. Prices of individual equities do not move in isolation; they respond to common macroeconomic shocks, sectoral rotations, and cross-asset contagion. Viewed through this lens, the market begins to resemble a many-body system in statistical mechanics: a collection of “particles” (assets) whose local states (returns) are coupled through an interaction structure that may itself evolve with time.

The objective of this project is to exploit that analogy rigorously. We seek a model that simultaneously captures three empirical regularities of equity markets:

- (i) **Cross-sectional dependence.** Stocks move together, but not uniformly; the correlation structure is sparse and time-varying.
- (ii) **Discrete collective behaviour.** Markets appear to shift between qualitatively different phases—bullish calm, bearish stress, and transitional uncertainty—in which the same news can produce very different price reactions.
- (iii) **Non-stationarity.** Any model with fixed parameters will eventually decay in predictive power as the underlying economic environment changes.

Our approach is to treat the market as a spin system. We encode daily returns into discrete states and model the joint distribution of those states through an energy-based Hamiltonian. The journey, however, is not immediate. We proceed through a deliberate sequence of four modelling stages, each correcting a specific limitation of its predecessor:

1. A *static Ising model* that treats the market as a collection of binary spins with fixed interactions. It is a natural starting point, but it cannot adapt to temporal change.

2. A *dynamic Ising idea* in which interactions themselves become stochastic processes. Conceptually appealing, it is computationally intractable because the partition function becomes a moving target.
3. A *conditional Ising model* that replaces the joint distribution with a lagged conditional logistic form. This restores tractability through rolling-window estimation and ℓ_1 regularisation, yet it still forces every day into an up-or-down state, discarding valuable information about low-activity periods.
4. A *Blume–Capel model* with tri-state spins $\{-1, 0, +1\}$, regime-switching hidden Markov dynamics, mean-field inference, pseudo-likelihood learning, and an online portfolio engine. This is the final and most expressive framework.

Before developing these models in detail, we first set up the statistical-mechanics language on which they rely. The next section introduces the Ising model, explains how its spins and interactions translate naturally into market variables, and establishes the notation used throughout the rest of the report.

2 Background and Model Foundations

2.1 The Ising Model in Statistical Mechanics

The Ising model was introduced by Lenz and solved in one dimension by Ising in the 1920s [1] to explain ferromagnetism. In its generalised form, one considers N particles indexed $i \in \{1, \dots, N\}$, each taking a spin state $\sigma_i \in \{+1, -1\}$. The configuration $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N)$ is assigned a probability governed by the Hamiltonian energy function

$$H(\boldsymbol{\sigma}; \mathbf{W}, \mathbf{h}) = - \sum_{i < j} W_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i, \quad (1)$$

where $\mathbf{W} \in \mathbb{R}^{N \times N}$ is a symmetric interaction matrix with $W_{ii} = 0$, and $\mathbf{h} \in \mathbb{R}^N$ is an external field vector. The Gibbs distribution is then

$$P(\boldsymbol{\sigma}) = \frac{1}{Z(\mathbf{W}, \mathbf{h})} \exp(-H(\boldsymbol{\sigma}; \mathbf{W}, \mathbf{h})), \quad Z(\mathbf{W}, \mathbf{h}) = \sum_{\mathbf{x} \in \{-1, +1\}^N} \exp(-H(\mathbf{x}; \mathbf{W}, \mathbf{h})). \quad (2)$$

The partition function Z is the normalising constant. It is deceptively simple: for N spins, the sum ranges over 2^N states, rendering exact computation infeasible for $N \gtrsim 30$.

2.2 From Magnets to Markets

To transplant the Ising model into finance, we identify each spin with the daily directional return of a stock. Let $P_t^{(i)} > 0$ denote the adjusted closing price of asset i at discrete trading time $t \in \mathcal{T} = \{0, 1, \dots, T\}$. The simple return is

$$r_t^{(i)} := \frac{P_t^{(i)} - P_{t-1}^{(i)}}{P_{t-1}^{(i)}}. \quad (3)$$

We encode this into a binary spin by thresholding at zero:

$$X_{i,t} = \begin{cases} +1 & \text{if } r_t^{(i)} \geq 0, \\ -1 & \text{if } r_t^{(i)} < 0. \end{cases} \quad (4)$$

The cross-sectional configuration at time t is $\mathbf{X}_t = (X_{1,t}, \dots, X_{N,t}) \in \{-1, +1\}^N$. Viewed as a stochastic process $\{\mathbf{X}_t\}_{t \geq 0}$, the market becomes a time-indexed sequence of spin configurations. The analogue of the interaction matrix $\mathbf{W}$ is then a network of cross-asset dependencies: if $W_{ij} > 0$, stocks i and j tend to move in the same direction; if $W_{ij} < 0$, they tend to oppose each other.

3 The Static Ising Model

3.1 Binary Market Representation

The most direct application of the Ising framework to market data is to assume a single, time-homogeneous interaction structure. We posit

$$P(\mathbf{X}_t = \mathbf{x}) \propto \exp\left(\frac{1}{2}\mathbf{x}^\top \mathbf{W} \mathbf{x} + \mathbf{h}^\top \mathbf{x}\right), \quad (5)$$

with fixed parameters $(\mathbf{W}, \mathbf{h})$ estimated over the entire history. This is the static Ising baseline.

What does this capture? The quadratic term $\mathbf{x}^\top \mathbf{W} \mathbf{x}$ encodes pairwise cross-sectional dependence. A positive entry W_{ij} rewards configurations in which $X_{i,t}$ and $X_{j,t}$ agree, while the linear term h_i captures an asset-specific directional bias—the analogue of an external magnetic field tilting a spin upward or downward regardless of its neighbours.

3.2 Limitations

The limitation of (5) is immediate and severe: the structure remains static across time. Financial interaction networks are not geological formations; they shift with monetary policy, earnings cycles, and geopolitical events. A fixed $\mathbf{W}$ estimated in 2018 will almost surely misrepresent the dependencies that prevailed during the COVID-19 crash of 2020 or the post-pandemic recovery. Moreover, the model as stated lacks any temporal covariance structure; it is a sequence of independent snapshots rather than a dynamical system.

These shortcomings motivate the next step: allowing $(\mathbf{W}_t, \mathbf{h}_t)$ to evolve.

4 The Dynamic Ising Idea

4.1 Motivation for Time-Varying Interactions

If market structure changes, the model parameters should change with it. The conceptual extension is to replace $(\mathbf{W}, \mathbf{h})$ with a stochastic process $(\mathbf{W}_t, \mathbf{h}_t)_{t \geq 0}$:

$$P(\mathbf{X}_t = \mathbf{x} \mid \mathbf{W}_t, \mathbf{h}_t) \propto \exp\left(\frac{1}{2}\mathbf{x}^\top \mathbf{W}_t \mathbf{x} + \mathbf{h}_t^\top \mathbf{x}\right). \quad (6)$$

This mitigates the static-nature problem and, in principle, captures temporal dependence through the joint law of $\{(\mathbf{W}_t, \mathbf{h}_t, \mathbf{X}_t)\}$.

4.2 Why Direct Time-Series Modelling Is Difficult

Despite its conceptual appeal, the dynamic Ising formulation poses two formidable obstacles:

Computational intractability. The partition function at time t is

$$Z(\mathbf{W}_t, \mathbf{h}_t) = \sum_{\mathbf{x} \in \{-1, +1\}^N} \exp\left(\frac{1}{2} \mathbf{x}^\top \mathbf{W}_t \mathbf{x} + \mathbf{h}_t^\top \mathbf{x}\right), \quad (7)$$

which must be recomputed whenever the parameters change. Without additional structure on the evolution of $(\mathbf{W}_t, \mathbf{h}_t)$, one cannot factorise the likelihood across time, and inference collapses to a sequence of intractable sums.

Unspecified parameter dynamics. Equation (6) states *what* the model is at each t , but not *how* $(\mathbf{W}_t, \mathbf{h}_t)$ moves. One could impose a vector autoregression or a diffusion on the vectorised parameters, but the dimension of $\mathbf{W}_t$ is $O(N^2)$. For $N = 50$, this means tracking 2,500 interaction coefficients plus 50 fields—a high-dimensional latent state whose own dynamics would require heroic assumptions and fragile estimation.

Rather than specifying the evolution of $(\mathbf{W}_t, \mathbf{h}_t)$ directly, we therefore pivot to a more tractable approach: conditioning on the past.

5 The Conditional Ising Model

5.1 Lagged Conditional Dependence

We modify the distribution of $\mathbf{X}_t$ by making it depend explicitly on the previous configuration $\mathbf{X}_{t-1}$. The conditional probability is

$$P(\mathbf{X}_t = \mathbf{x} \mid \mathbf{X}_{t-1}) \propto \exp(\mathbf{x}^\top \mathbf{W} \mathbf{X}_{t-1} + \mathbf{h}^\top \mathbf{x}). \quad (8)$$

The key intuition is that interaction effects appear with a one-period lag: today's return on stock i is influenced by yesterday's returns on other stocks $j \neq i$, but *not* by today's contemporaneous moves. A crucial implication follows immediately: conditional on $\mathbf{X}_{t-1}$, the components $X_{i,t}$ become mutually independent. The joint distribution factorises into a product of univariate logistic probabilities.

5.2 Logistic Conditional Structure

For each stock i at time t , the conditional probability of an upward move is

$$P(X_{i,t} = 1 \mid \mathbf{X}_{t-1}) = \sigma(2\eta_{i,t}), \quad \eta_{i,t} = h_i + \sum_{j \neq i} W_{ij} X_{j,t-1}, \quad (9)$$

where $\sigma(u) = 1/(1 + e^{-u})$ is the logistic sigmoid. Equivalently,

$$\text{logit}(P(X_{i,t} = 1 \mid \mathbf{X}_{t-1})) = 2 \left(\sum_{j \neq i} W_{ij} X_{j,t-1} + h_i \right). \quad (10)$$

Here, the interaction matrix $\mathbf{W}$ is symmetric and has zeros along its diagonal, so $W_{ij} = W_{ji}$ and $W_{ii} = 0$ for every asset i .

5.3 Rolling-Window Estimation

To accommodate non-stationarity without modelling the parameter dynamics explicitly, we adopt a rolling-window strategy. At each time t , we define a local training window

$$\mathcal{W}_t = \{t - L + 1, \dots, t\}, \quad (11)$$

of fixed length L , and assume approximate local stationarity within $\mathcal{W}_t$. The parameters $(\mathbf{W}(t), \mathbf{h}(t))$ are re-estimated from the data in $\mathcal{W}_t$ alone.

For a fixed window, the conditional log-likelihood is

$$\ell_t(\mathbf{W}(t), \mathbf{h}(t)) = \sum_{s=t-L+1}^t \sum_{i=1}^N \left[X_{i,s} \eta_{i,s}(t) - \log(2 \cosh(\eta_{i,s}(t))) \right], \quad (12)$$

where $\eta_{i,s}(t) = h_i(t) + \sum_{j \neq i} W_{ij}(t) X_{j,s-1}$.

5.4 ℓ_1 -Regularised Estimation and Coordinate Descent

The number of interaction parameters grows quadratically with N . To obtain a sparse, interpretable network and prevent overfitting in short windows, we impose an ℓ_1 -penalty on the off-diagonal entries of $\mathbf{W}(t)$. The resulting conditional specification may be viewed as a modified vector autoregressive model of order one, where the dependence structure is governed by a logistic interaction mechanism rather than a linear Gaussian one. The penalised objective is (Ravikumar P., Wainwright M. J., Lafferty J. D. [5]),

$$(\hat{\mathbf{W}}(t), \hat{\mathbf{h}}(t)) = \arg \max_{\mathbf{W}, \mathbf{h}} \left\{ \ell_t(\mathbf{W}, \mathbf{h}) - \lambda \sum_{i=1}^N \|\mathbf{W}_{i,\cdot}\|_1 \right\}. \quad (13)$$

Because the penalised objective is convex but non-smooth, we optimise via coordinate descent: coefficients are updated one at a time while holding all others fixed (Friedman J. H., Hastie T., Tibshirani R. [6]). The ℓ_1 term induces soft-thresholding behaviour, pushing weak interactions exactly to zero and retaining only the strongest edges. The resulting $\hat{\mathbf{W}}(t)$ need not be symmetric; we symmetrise it post-estimation via

$$\hat{\mathbf{W}}_{\text{sym}}(t) = \frac{\hat{\mathbf{W}}(t) + \hat{\mathbf{W}}(t)^\top}{2}. \quad (14)$$

Algorithm 1 Rolling-Window ℓ_1 -Regularised Conditional Ising Estimation**Require:** Binary spin sequence $\{\mathbf{X}_t\}$, window length L , penalty $\lambda > 0$, tolerance ε

```

1: for  $t = L, L + 1, \dots, T$  do
2:    $\mathcal{W}_t \leftarrow \{t - L + 1, \dots, t\}$ 
3:   Initialise  $\mathbf{W}^{(0)}, \mathbf{h}^{(0)}$ 
4:   repeat
5:     for  $i = 1, \dots, N$  do
6:       for  $j \neq i$  do
7:         Update  $W_{ij}$  by univariate Newton step with soft-thresholding
8:       end for
9:       Update  $h_i$  by univariate Newton step
10:    end for
11:  until objective change  $< \varepsilon$ 
12:   $\hat{\mathbf{W}}_{\text{sym}}(t) \leftarrow \frac{1}{2}(\hat{\mathbf{W}}(t) + \hat{\mathbf{W}}(t)^\top)$ 
13:  Store  $(\hat{\mathbf{W}}_{\text{sym}}(t), \hat{\mathbf{h}}(t))$ 
14: end for

```

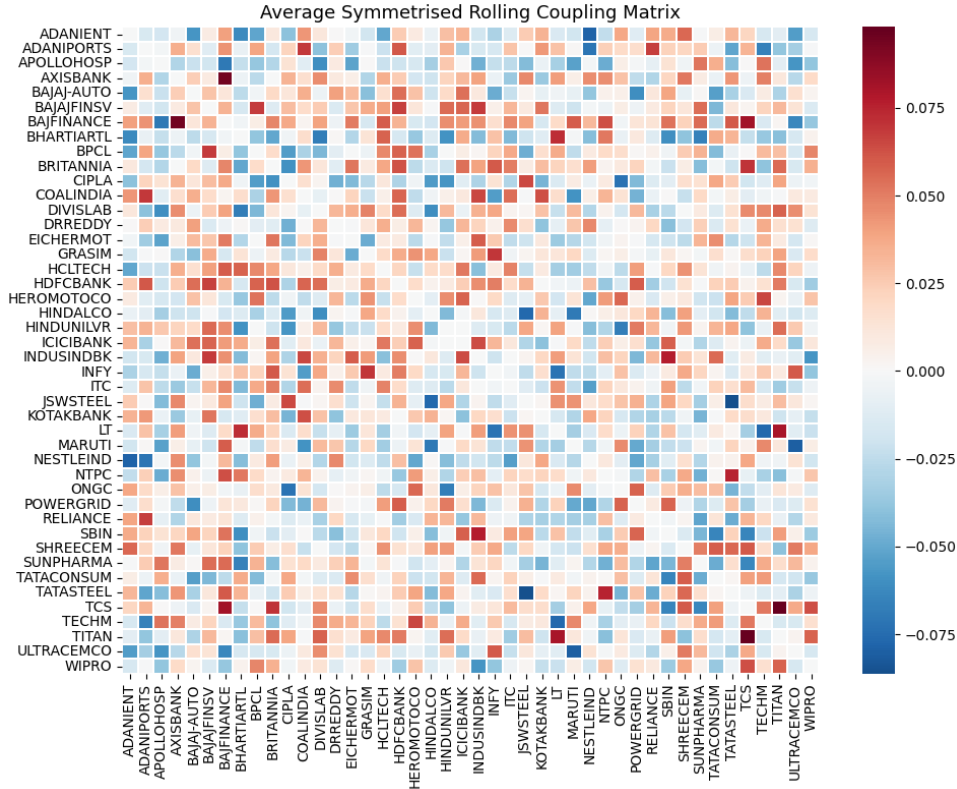


Figure 1: Estimated Interaction Matrix Heatmap

5.5 Prediction and Empirical Performance

Given the estimated parameters at time t , the one-step-ahead forecast for stock i is

$$\hat{\eta}_{i,t+1}(t) = \hat{h}_i(t) + \sum_{j \neq i} \hat{W}_{ij}(t) X_{j,t}, \quad \hat{X}_{i,t+1} = \begin{cases} +1 & \text{if } \hat{\eta}_{i,t+1}(t) \geq 0, \\ -1 & \text{otherwise.} \end{cases} \quad (15)$$

We evaluated this rolling conditional Ising model on the Nifty 50 constituents (as of 1 January, 2025) over the period 17 November 2017 to 1 January 2025, using a rolling window of $L = 30$ trading days. The directional accuracy was approximately 50.50% for positive predictions and 49.70% for negative predictions, yielding precision 0.5114, recall 0.5050, and an F1 score of 0.5082. The confusion matrix is shown in table 1.

Table 1: Confusion matrix for the rolling conditional Ising model on Nifty 50 constituents.

	Pred. Down	Pred. Up
True Down	27,852	28,185
True Up	28,910	29,498

The estimated interaction matrices (fig. 1) exhibit sparse, structured patterns with pronounced sectoral clusters—financials interacting with financials, energy with energy—but the predictive performance is essentially at the level of a biased coin. The model is better than the static baseline because it adapts locally, yet it still falls short. Why?

6 Why Binary Spins Are Not Enough

The Achilles’ heel of the conditional Ising model is its binary encoding. By forcing every day into either $+1$ (up) or -1 (down), we discard all information about the *magnitude* of the move. In reality, a large fraction of daily returns cluster near zero. Treating a $+0.05\%$ drift and a $+3\%$ rally as identical states, while labelling a -0.01% tick as the opposite of a $+0.01\%$ tick, injects noise into the spin configuration.

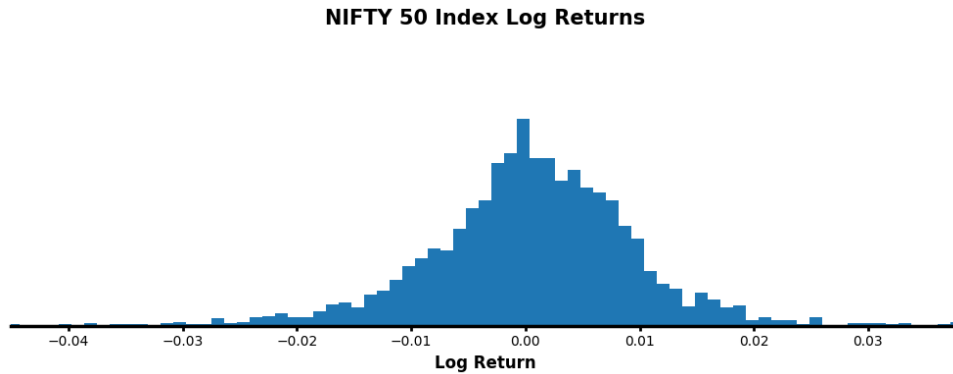


Figure 2: Nifty 50 log-returns

This is visible in the empirical distribution of Nifty 50 log returns (fig. 2), which displays a sharp central peak around zero with wide, thin tails. The binary model squashes that central mass into two opposing states, destroying the distinction between low-activity and high-activity days. A natural remedy is to introduce a *neutral* state, allowing the model to recognise periods of directional ambiguity. This leads directly to the Blume–Capel framework.

7 The Blume–Capel Model

7.1 Tri-State Representation

The Blume–Capel model ([2],[3]) generalises the Ising framework by allowing each spin to take three states: $s_i \in \mathcal{S} = \{-1, 0, +1\}$. In our context,

- $+1$ represents a positive, meaningfully large movement;
- 0 represents a neutral or low-activity movement;
- -1 represents a negative, meaningfully large movement.

The market configuration at time t is $\mathbf{X}_t = (X_{1,t}, \dots, X_{N,t}) \in \mathcal{S}^N$.

7.2 Hamiltonian and Physical Interpretation

The Blume–Capel Hamiltonian is

$$H(\mathbf{x}; \boldsymbol{\theta}) = -\sum_{i=1}^N h_i x_i - \sum_{i=1}^N d_i x_i^2 - \sum_{1 \leq i < j \leq N} J_{ij} x_i x_j, \quad (16)$$

where the parameter vector is $\boldsymbol{\theta} = (\mathbf{h}, \mathbf{d}, \mathbf{J})$ with:

- $\mathbf{h} \in \mathbb{R}^N$: the linear field, analogous to the external magnetic field in the Ising model, capturing directional bias;
- $\mathbf{d} \in \mathbb{R}^N$: the crystal field, a quadratic single-ion anisotropy that controls the *activity* of a spin—how likely it is to be in state 0 versus ± 1 ;
- $\mathbf{J} \in \mathbb{R}^{N \times N}$: the exchange coupling matrix, symmetric with $J_{ii} = 0$, capturing pairwise cross-asset interactions.

The corresponding Gibbs distribution is

$$P(\mathbf{X}_t = \mathbf{x} \mid \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp(-H(\mathbf{x}; \boldsymbol{\theta})), \quad Z(\boldsymbol{\theta}) = \sum_{\mathbf{x} \in \mathcal{S}^N} \exp(-H(\mathbf{x}; \boldsymbol{\theta})). \quad (17)$$

Remark 7.1 (Not a Potts model). The Blume–Capel model is structurally distinct from the q -state Potts model. The Potts Hamiltonian uses Kronecker-delta coupling $\delta(s_i, s_j)$ with no notion of “opposite” states or a neutral resting state. The crystal-field term $d_i x_i^2$ has no analogue in the Potts framework.

The crystal field is particularly important financially. When $d_i \ll 0$, the term $-d_i x_i^2$ penalises active states, pushing the spin toward 0; this corresponds to a calm, low-volatility market. When $d_i \gg 0$, the spin is encouraged to be active (± 1), corresponding to a high-volatility, directional regime. The interaction matrix $\mathbf{J}$ then determines whether those active spins align (ferromagnetic, trending market) or oppose one another (frustrated, rotational market).

8 PCA and Latent Spin Encoding

Directly modelling interactions between all N assets leads to a very high-dimensional system with $O(N^2)$ coupling parameters. Moreover, raw returns are noisy. To obtain a lower-dimensional, more stable latent representation, we apply Principal Component Analysis (PCA) inside each rolling training window.

8.1 Training Window and SVD

At walk-forward step t , the training window of length W is

$$\mathcal{W}_t = \{\mathbf{r}_{t-W+1}, \dots, \mathbf{r}_t\} \subset \mathbb{R}^N. \quad (18)$$

We organise these as rows of a data matrix $\mathbf{R}_t \in \mathbb{R}^{W \times N}$. After centering,

$$\boldsymbol{\mu}_t = \frac{1}{W} \mathbf{R}_t^\top \mathbf{1}_W, \quad \tilde{\mathbf{R}}_t = \mathbf{R}_t - \mathbf{1}_W \boldsymbol{\mu}_t^\top, \quad (19)$$

we compute the singular value decomposition $\tilde{\mathbf{R}}_t = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top$. Let $\mathbf{P}_t = \mathbf{V}_{:,1:n} \in \mathbb{R}^{N \times n}$ denote the loadings matrix of the leading n principal directions, where n is chosen to capture a target fraction $v^* \in (0, 1)$ of total variance.

The projection of a return vector $\mathbf{r} \in \mathbb{R}^N$ into PC space is

$$\mathbf{z} = \mathbf{P}_t^\top (\mathbf{r} - \boldsymbol{\mu}_t) \in \mathbb{R}^n. \quad (20)$$

Conversely, a signal $\mathbf{s} \in \mathbb{R}^n$ maps back to asset space via $\hat{\mathbf{r}} = \mathbf{P}_t \mathbf{s}$.

Algorithm 2 PCA Preprocessing and Spin Encoding

Require: Return matrix $\mathbf{R} \in \mathbb{R}^{W \times N}$, target variance v^* , min components $n_{\min}$, max $n_{\max}$

- 1: $\boldsymbol{\mu} \leftarrow \frac{1}{W} \mathbf{R}^\top \mathbf{1}_W$
 - 2: $\tilde{\mathbf{R}} \leftarrow \mathbf{R} - \mathbf{1}_W \boldsymbol{\mu}^\top$
 - 3: Compute SVD: $\tilde{\mathbf{R}} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top$ with singular values $\sigma_1 \geq \dots \geq \sigma_{\min(W,N)}$
 - 4: $s_k \leftarrow \sigma_k^2 / \sum_j \sigma_j^2$ ▷ Explained variance ratio
 - 5: $c_k \leftarrow \sum_{j=1}^k s_j$ ▷ Cumulative variance
 - 6: $n \leftarrow \min\{k : c_k \geq v^*\} + 1$
 - 7: $n \leftarrow \text{clip}(n, n_{\min}, \min(n_{\max}, N - 1))$
 - 8: $\mathbf{P} \leftarrow \mathbf{V}_{:,1:n}$
 - 9: **return** $(\boldsymbol{\mu}, \mathbf{P}, \{s_k\}_{k=1}^n, n)$
-

8.2 Quantile and Z-Score Thresholding

The Blume–Capel model requires discrete spins. We map PC scores $\mathbf{z} \in \mathbb{R}^n$ to spins $\mathbf{s} \in \{-1, 0, +1\}^n$ via per-component thresholds.

Quantile encoding. For the j -th PC component, let $\{z_{1,j}, \dots, z_{W,j}\}$ be the training scores. Given a quantile parameter $q \in (0, \frac{1}{3}]$, define

$$\tau_j^{\text{lo}} = Q_q(\{z_{t,j}\}), \quad \tau_j^{\text{hi}} = Q_{1-q}(\{z_{t,j}\}), \quad (21)$$

where $Q_\alpha(\cdot)$ denotes the empirical α -quantile. The spin encoding is

$$s_j = \begin{cases} -1 & \text{if } z_j < \tau_j^{\text{lo}}, \\ 0 & \text{if } \tau_j^{\text{lo}} \leq z_j \leq \tau_j^{\text{hi}}, \\ +1 & \text{if } z_j > \tau_j^{\text{hi}}. \end{cases} \quad (22)$$

9 Inference and Learning

9.1 Intractability of Exact Inference

The normalising constant of the Blume–Capel model is

$$Z(\boldsymbol{\theta}) = \sum_{\mathbf{x} \in \{-1, 0, +1\}^N} \exp(-H(\mathbf{x}; \boldsymbol{\theta})). \quad (23)$$

Since the state space has cardinality 3^N , exact computation is infeasible even for moderate N . This makes the exact likelihood, exact marginals, and exact parameter learning computationally intractable. We therefore require two complementary approximations: mean-field inference for the partition function, and pseudo-likelihood estimation for parameter learning.

9.2 Mean-Field Approximation

We employ the naïve mean-field (variational) approximation, replacing the true joint distribution by a fully factorised variational family:

$$q(\mathbf{x}) = \prod_{i=1}^N q_i(x_i), \quad q_i(x_i) = \sum_{a \in \mathcal{S}} p_i(a) \cdot \mathbf{1}[x_i = a], \quad (24)$$

with variational parameters $\{p_i(a)\}$ satisfying $p_i(a) \geq 0$ and $\sum_a p_i(a) = 1$.

Definition 9.1 (Effective Field). Under mean-field, the effective field at node i is

$$h_i^{\text{eff}} = h_i + \sum_{j \neq i} J_{ij} m_j, \quad (25)$$

where $m_j = \mathbb{E}_q[X_j] = p_j(+1) - p_j(-1)$ is the local magnetisation.

Proposition 9.1 (Mean-Field Marginals). *For node i with effective field h_i^{eff} , the unnormalised log-probabilities are*

$$\log \tilde{p}_i(-1) = d_i - h_i^{\text{eff}}, \quad \log \tilde{p}_i(0) = 0, \quad \log \tilde{p}_i(+1) = d_i + h_i^{\text{eff}}. \quad (26)$$

Normalising via the log-sum-exp trick with $\ell_i^{\max} = \max\{d_i + h_i^{\text{eff}}, d_i - h_i^{\text{eff}}, 0\}$,

$$Z_i = \exp(\ell_i^{\max}) \left[\exp(d_i + h_i^{\text{eff}} - \ell_i^{\max}) + \exp(d_i - h_i^{\text{eff}} - \ell_i^{\max}) + \exp(-\ell_i^{\max}) \right], \quad (27)$$

and the probabilities are $p_i(a) = \exp(\log \tilde{p}_i(a) - \ell_i^{\max}) / Z_i$. The sufficient statistics are

$$m_i = \mathbb{E}_q[X_i] = p_i(+1) - p_i(-1) \in (-1, +1), \quad q_i^{(2)} = \mathbb{E}_q[X_i^2] = p_i(+1) + p_i(-1) = 1 - p_i(0) \in [0, 1]. \quad (28)$$

Proof. The mean-field free energy is

$$\mathcal{F}[q] = \mathbb{E}_q[H(\mathbf{X})] + \sum_{\mathbf{x}} q(\mathbf{x}) \log q(\mathbf{x}). \quad (29)$$

Under the factorised ansatz, the energy term becomes

$$\mathbb{E}_q[H(\mathbf{X})] = - \sum_i h_i m_i - \sum_i d_i q_i^{(2)} - \sum_{i < j} J_{ij} m_i m_j. \quad (30)$$

Taking the functional derivative $\delta \mathcal{F} / \delta q_i(a)$ subject to $\sum_a q_i(a) = 1$ yields

$$-h_i a - d_i a^2 - \sum_{j \neq i} J_{ij} m_j a + \log q_i(a) + 1 + \lambda = 0, \quad (31)$$

where λ is a Lagrange multiplier. Solving gives $q_i(a) \propto \exp(h_i^{\text{eff}} a + d_i a^2)$. For $a \in \{-1, 0, +1\}$, this produces the stated unnormalised probabilities. The log-sum-exp formulation ensures numerical stability by preventing exponent overflow. $\square$

These quantities satisfy a self-consistent fixed-point relation, since h_i^{eff} depends on $\{m_j\}$, which in turn depend on h_i^{eff} . The local magnetisation m_i measures directional bias, while the local activity $q_i^{(2)}$ measures how often the spin departs from the neutral state.

Algorithm 3 Mean-Field Fixed-Point Iteration

Require: Parameters $(\mathbf{h}, \mathbf{d}, \mathbf{J})$, tolerance $\varepsilon > 0$, max iterations $I_{\max}$

```

1:  $\mathbf{m}^{(0)} \leftarrow \mathbf{0}_N$ 
2: for  $\tau = 1, 2, \dots, I_{\max}$  do
3:    $\mathbf{h}^{\text{eff}} \leftarrow \mathbf{h} + \mathbf{J}\mathbf{m}^{(\tau-1)}$ 
4:   Compute  $(p_i(+1), p_i(-1), p_i(0), \mathbf{m}^{(\tau)}, \mathbf{q}^{(2)})$  via proposition 9.1
5:   if  $\|\mathbf{m}^{(\tau)} - \mathbf{m}^{(\tau-1)}\|_{\infty} < \varepsilon$  then
6:     break
7:   end if
8: end for
9: return  $(\mathbf{m}^{(\tau)}, \mathbf{q}^{(2)})$ 
```

The proof of convergence using the Contraction Mapping Theorem can be found in the appendix.

9.3 Parameter Identifiability and the d -Centering Constraint

Proposition 9.2 (Joint Identifiability of $(\mathbf{h}, \mathbf{d})$). *Consider the transformation $h_i \mapsto h_i + \lambda$, $d_i \mapsto d_i - \lambda$ for all i and some $\lambda \in \mathbb{R}$. The marginals transform as:*

$$p_i(+1) \propto \exp((d_i - \lambda) + (h_i^{\text{eff}} + \lambda)) = \exp(d_i + h_i^{\text{eff}}) \quad (\text{invariant}), \quad (32)$$

$$p_i(-1) \propto \exp((d_i - \lambda) - (h_i^{\text{eff}} + \lambda)) = \exp(d_i - h_i^{\text{eff}} - 2\lambda) \quad (\text{not invariant}), \quad (33)$$

$$p_i(0) \propto 1 \quad (\text{invariant}). \quad (34)$$

Since $p_i(-1)$ changes, the full distribution is not invariant. Therefore, $(\mathbf{h}, \mathbf{d})$ are jointly identifiable.

A milder numerical issue is gradient saturation: when $|d_i| \gg 0$, the activity $q_i^{(2)}$ saturates near 0 or 1, causing the pseudo-likelihood gradient $\partial \text{PLL} / \partial d_i = x_i^2 - q_i^{(2)}$ to vanish. We address this by ℓ_2 regularisation on $\mathbf{d}$, gradient clipping, and a post-update d -centering step $\mathbf{d} \leftarrow \mathbf{d} - \bar{d} \mathbf{1}_N$ where $\bar{d} = \frac{1}{N} \sum_i d_i$. This is a mean-zero constraint for numerical conditioning, not a gauge fixing.

9.4 Pseudo-Likelihood Estimation

Since exact likelihood maximisation is infeasible, we use the pseudo-log-likelihood (PLL). For an observed spin configuration $\mathbf{x} \in \mathcal{S}^N$, the PLL of node i is the conditional log-probability given all other spins:

$$\text{PLL}_i(\mathbf{x}; \boldsymbol{\theta}) := \log p(x_i \mid \mathbf{x}_{\setminus i}; \boldsymbol{\theta}) = \log \frac{\exp(h_i^{\text{eff}}(\mathbf{x}_{\setminus i})x_i + d_i x_i^2)}{Z_i(\mathbf{x}_{\setminus i})}, \quad (35)$$

where $h_i^{\text{eff}}(\mathbf{x}_{\setminus i}) = h_i + \sum_{j \neq i} J_{ij} x_j$. The total PLL is

$$\text{PLL}(\mathbf{x}; \boldsymbol{\theta}) = \sum_{i=1}^N \text{PLL}_i(\mathbf{x}; \boldsymbol{\theta}). \quad (36)$$

Theorem 9.1 (PLL Gradient Formulae). *The gradients of the total PLL with respect to the parameters are:*

$$\frac{\partial \text{PLL}}{\partial h_i} = x_i - m_i(\mathbf{x}_{\setminus i}) = x_i - \mathbb{E}_{p(\cdot \mid \mathbf{x}_{\setminus i})}[X_i], \quad (37)$$

$$\frac{\partial \text{PLL}}{\partial d_i} = x_i^2 - q_i^{(2)}(\mathbf{x}_{\setminus i}) = x_i^2 - \mathbb{E}_{p(\cdot \mid \mathbf{x}_{\setminus i})}[X_i^2], \quad (38)$$

$$\frac{\partial \text{PLL}}{\partial J_{ij}} = x_i x_j - m_i m_j + (x_i - x_j)(m_i - m_j). \quad (39)$$

Here m_i and $q_i^{(2)}$ are the mean-field magnetisation and activity computed with $\mathbf{x}$ as the neighbour configuration.

Proof Sketch. For (9.1), note that $\text{PLL}_i = h_i^{\text{eff}} x_i + d_i x_i^2 - \log Z_i$. Differentiating,

$$\frac{\partial \text{PLL}_i}{\partial h_i} = x_i - \frac{\partial \log Z_i}{\partial h_i} = x_i - \mathbb{E}[X_i \mid \mathbf{x}_{\setminus i}] = x_i - m_i. \quad (40)$$

Equation for d_i follows identically with x_i^2 replacing x_i . For J_{ij} , the chain rule gives $\partial h_i^{\text{eff}}/\partial J_{ij} = x_j$ and $\partial h_j^{\text{eff}}/\partial J_{ij} = x_i$. Symmetrising ensures $\partial \text{PLL}/\partial J_{ij} = \partial \text{PLL}/\partial J_{ji}$.

Remark 9.1. The PLL is *not* a proper scoring rule. It conditions on observed neighbours rather than the true joint, so it cannot be used for model comparison across different model classes. However, its gradients are consistent estimators of the true parameter gradients (Hyvärinen, 2006), and we use it exclusively for parameter updates, not for scoring.

9.5 Mean-Field Log-Partition Approximation

For hidden Markov model emission probabilities, we require $p(\mathbf{x} \mid \boldsymbol{\theta})$. The exact partition function is intractable; we use the mean-field approximation.

Proposition 9.3 (Mean-Field Free Energy). *Let $\mathbf{m}^*$ be the fixed-point magnetisation from algorithm 3. The mean-field approximation to $\log Z$ is*

$$\log Z_{\text{MF}} = \sum_{i=1}^N \log Z_i(h_i^{\text{eff}}(\mathbf{m}^*)) - \sum_{i < j} J_{ij} m_i^* m_j^*, \quad (41)$$

where $Z_i(h) = 1 + 2 \exp(d_i) \cosh(h)$ is the single-site partition function. The mean-field log-emission is then

$$\log p_{\text{MF}}(\mathbf{x} \mid \boldsymbol{\theta}) = \mathbf{h}^\top \mathbf{x} + \mathbf{d}^\top (\mathbf{x} \odot \mathbf{x}) + \sum_{i < j} J_{ij} x_i x_j - \log Z_{\text{MF}}. \quad (42)$$

Proof. The mean-field variational upper bound on $\log Z$ is $\log Z \leq -\mathcal{F}[q^*]$, where q^* is the optimal factorised distribution. At the fixed point, the entropy term $-\sum_i \mathbb{E}_{q_i}[\log q_i]$ cancels with the excess in the energy term, leaving the stated expression. $\square$

10 Regime-Switching Extension

10.1 Hidden Regimes and HMM Formulation

Stable markets exhibit weak interactions, while crises induce strong collective motion. We therefore classify the market into K latent regimes and allow the Blume–Capel parameters to be regime-specific. Let $C_t \in \mathcal{C} = \{1, \dots, K\}$ denote the latent market regime at time t . We assume $\{C_t\}$ is a discrete-time Markov chain with transition matrix $\mathbf{A} \in \mathbb{R}^{K \times K}$, where $A_{ij} = P(C_t = j \mid C_{t-1} = i)$, and initial distribution $\boldsymbol{\pi} \in \Delta^{K-1}$.

Conditional on regime $C_t = k$, the spin configuration follows a regime-specific Blume–Capel distribution:

$$p(\mathbf{X}_t \mid C_t = k) = p_{\text{BC}}(\mathbf{X}_t; \boldsymbol{\theta}^{(k)}), \quad \boldsymbol{\theta}^{(k)} = (\mathbf{h}^{(k)}, \mathbf{d}^{(k)}, \mathbf{J}^{(k)}). \quad (43)$$

Marginally, $\mathbf{X}_t$ follows a mixture:

$$p(\mathbf{X}_t) = \sum_{k=1}^K P(C_t = k) p_{\text{BC}}(\mathbf{X}_t; \boldsymbol{\theta}^{(k)}). \quad (44)$$

To encourage regime persistence—markets do not flip from bull to bear overnight—we employ a sticky transition prior:

$$A_{ij} = \begin{cases} \alpha & \text{if } i = j, \\ \frac{1-\alpha}{K-1} & \text{if } i \neq j, \end{cases} \quad (45)$$

with persistence parameter $\alpha \in (0, 1)$ (default $\alpha = 0.70$). This is a convex combination of the identity and uniform matrices: $\mathbf{A} = \alpha \mathbf{I} + \frac{1-\alpha}{K} \mathbf{1}\mathbf{1}^\top$.

10.2 Log-Space Forward Filtering

Definition 10.1 (Forward Messages). The forward message $\alpha_t \in \Delta^{K-1}$ is the filtered regime posterior:

$$\alpha_{t,k} = P(C_t = k \mid \mathbf{X}_{1:t}) = \frac{p(\mathbf{X}_t \mid C_t = k) \sum_{i=1}^K \alpha_{t-1,i} A_{ik}}{\sum_j p(\mathbf{X}_t \mid C_t = j) \sum_i \alpha_{t-1,i} A_{ij}}. \quad (46)$$

Theorem 10.1 (Log-Space Forward Recursion). Define $\log \alpha_t$ element-wise. The numerically stable recursion is

$$\log \tilde{\alpha}_{t,k} = \text{logsumexp}_{i=1}^K (\log A_{ik} + \log \alpha_{t-1,i}) + \frac{\log e_{t,k}}{\tau}, \quad (47)$$

$$\log \alpha_{t,k} = \log \tilde{\alpha}_{t,k} - \text{logsumexp}_{k=1}^K (\log \tilde{\alpha}_t), \quad (48)$$

where $e_{t,k} = p(\mathbf{X}_t \mid C_t = k; \theta^{(k)})$ is the emission probability and $\tau \geq 1$ is the emission temperature.

Proof. Equation (38) follows from taking logs of the prediction step $\tilde{\alpha}_{t,k} = \sum_i \alpha_{t-1,i} A_{ik} \cdot e_{t,k}^{1/\tau}$. The log-sum-exp formulation avoids underflow: when $e_{t,k} \sim 10^{-300}$, $\log e_{t,k} \sim -690$, which is perfectly representable in double precision, whereas $e_{t,k}$ underflows to zero in linear space. $\square$

Remark 10.1 (Emission Temperature). The temperature $\tau \geq 1$ flattens emission log-probabilities. As $\tau \rightarrow \infty$, all emissions become equal and the filter reduces to the prior. This is crucial near criticality in the mean-field approximation, where small errors in $\theta^{(k)}$ can produce wildly different emission probabilities. Typical range: $\tau \in [1.0, 3.0]$.

Algorithm 4 Log-Space Forward Filter Step

Require: Previous log-belief $\log \alpha_{t-1} \in \mathbb{R}^K$, emissions $\mathbf{e}_t \in \mathbb{R}_+^K$, transition $\mathbf{A} \in \mathbb{R}^{K \times K}$, temperature $\tau \geq 1$

- 1: $\log \mathbf{e}_t \leftarrow \log(\max(\mathbf{e}_t, 10^{-300}))/\tau$ $\triangleright$ Clip and temper
 - 2: $\log \mathbf{A} \leftarrow \log(\max(\mathbf{A}, 10^{-300}))$
 - 3: $\log \tilde{\alpha}_t \leftarrow \text{logsumexp}(\log \mathbf{A} + \log \alpha_{t-1}^\top, \text{axis} = 0) + \log \mathbf{e}_t$ $\triangleright$ Prediction + update
 - 4: $\log \alpha_t \leftarrow \log \tilde{\alpha}_t - \text{logsumexp}(\log \tilde{\alpha}_t)$ $\triangleright$ Normalise
 - 5: **return** $\exp(\log \alpha_t)$
-

10.3 Multi-Step Regime Prediction

Given current log-belief $\log \alpha_t$, the H -step ahead regime distribution is

$$\log \alpha_{t+H} = \underbrace{\text{logsumexp}(\log \mathbf{A} + \dots \text{logsumexp}(\log \mathbf{A} + \log \alpha_t^\top) \dots)}_{H \text{ times}}. \quad (49)$$

10.4 Modified EM-Style Learning

The complete training loop interleaves regime filtering, weighted pseudo-likelihood maximisation, and transition matrix updates.

Algorithm 5 RARSI Training (Online EM with Warmup)

Require: Spin sequence $\{\mathbf{x}_t\}_{t=1}^T$, passes P , patience p , warmup W_{up}

```

1: Initialise  $\{M_k\}_{k=1}^K$ ,  $\mathbf{A}$ ,  $\boldsymbol{\pi}$ 
2:  $\text{pll\_history} \leftarrow []$ 
3: for  $\text{pass} = 1, \dots, P$  do
4:   Reset HMM:  $\boldsymbol{\alpha}_0 \leftarrow \boldsymbol{\pi}$ 
5:    $\text{pass\_pll} \leftarrow 0$ 
6:   for  $t = 1, \dots, T$  do
7:      $\mathbf{e}_t \leftarrow [M_k.\text{emission}(\mathbf{x}_t)]_{k=1}^K$ 
8:      $\mathbf{w}_t \leftarrow \text{HMMFilter.step}(\mathbf{e}_t)$  ▷ Via algorithm 4
9:      $\mathbf{w}_t \leftarrow \max(\mathbf{w}_t, 0.01) / \sum_k w_{t,k}$  ▷ Floor and renormalise
10:    for  $k = 1, \dots, K$  do
11:       $\text{OMDUpdater3.step}(M_k, \mathbf{x}_t, w_{t,k})$  ▷ Via algorithm 6
12:    end for
13:     $\mathbf{A} \leftarrow \text{TransitionEG.step}(\mathbf{A}, \boldsymbol{\alpha}_{t-1}, \mathbf{e}_t)$  ▷ Via algorithm 7
14:     $\text{pass\_pll} \leftarrow \text{pass\_pll} + \sum_k w_{t,k} \cdot M_k.\log\_emission(\mathbf{x}_t)$ 
15:  end for
16:   $\text{avg\_pll} \leftarrow \text{pass\_pll} / T$ 
17:   $\text{pll\_history.append}(\text{avg\_pll})$ 
18:  if  $\text{pass} \geq W_{\text{up}}$  then
19:     $\text{prev\_avg} \leftarrow \text{mean}(\text{pll\_history}[-p-1 : -1])$ 
20:    if  $\text{avg\_pll} - \text{prev\_avg} < 10^{-4}$  then
21:       $\text{no\_improve} \leftarrow \text{no\_improve} + 1$ 
22:    else
23:       $\text{no\_improve} \leftarrow 0$ 
24:    end if
25:    if  $\text{no\_improve} \geq p$  then
26:      break
27:    end if
28:  end if
29: end for
30: return  $\{M_k\}, \mathbf{A}, \{\mathbf{w}_t\}$ 

```

Remark 10.2 (Convergence Guarantees). The training loop is not formal online EM (Cappé & Moulines, 2009) because: (i) we use SGD with soft labels rather than sufficient statistics accumulation; (ii) early in training, regime beliefs are near-uniform ($\mathbf{w}_t \approx \frac{1}{K}\mathbf{1}$), making gradients noisy; (iii) the PLL objective is non-convex in $(\mathbf{h}, \mathbf{d}, \mathbf{J})$ jointly. The warmup period W_{up} prevents false convergence during the initial random-walk phase. True online EM would require $O(KN^2)$ sufficient statistics storage, which is architecturally complex for this implementation.

11 Online Learning and Trading System

11.1 Weighted Pseudo-Likelihood Objective

At each time step, the HMM provides soft regime responsibilities $w_{t,k} = \alpha_{t,k}$. The weighted PLL objective for regime k is

$$\mathcal{L}_t^{(k)}(\boldsymbol{\theta}^{(k)}) = w_{t,k} \cdot \text{PLL}(\mathbf{x}_t; \boldsymbol{\theta}^{(k)}) - \Omega(\boldsymbol{\theta}^{(k)}), \quad (50)$$

where Ω is a composite regularisation functional.

Definition 11.1 (Composite Regularisation).

$$\Omega(\boldsymbol{\theta}) = \frac{\lambda_2}{2} \|\mathbf{h}\|_2^2 + \frac{\lambda_2^{(d)}}{2} \|\mathbf{d}\|_2^2 + \frac{\lambda_2}{2} \|\mathbf{J}\|_F^2 + \lambda_1^{(J)} \|\mathbf{J}\|_{1,\text{off-diag}}, \quad (51)$$

where $\lambda_2 > 0$ penalises $\mathbf{h}$ and $\mathbf{J}$ (prevents overfitting, ensures strong convexity), $\lambda_2^{(d)} > 0$ penalises $\mathbf{d}$ (prevents gradient saturation), and $\lambda_1^{(J)} > 0$ induces sparse coupling via ℓ_1 regularisation on off-diagonal J_{ij} .

11.2 Composite Objective and the Need for Proximal Methods

The weighted objective $\mathcal{L}_t^{(k)}$ is the sum of a differentiable (smooth) term and a non-differentiable (non-smooth) term. To see this clearly, drop the regime index k for brevity and write

$$\mathcal{L}_t(\boldsymbol{\theta}) = \underbrace{w_t \cdot \text{PLL}(\mathbf{x}_t; \boldsymbol{\theta}) - \frac{\lambda_2}{2} \|\mathbf{h}\|_2^2 - \frac{\lambda_2^{(d)}}{2} \|\mathbf{d}\|_2^2 - \frac{\lambda_2}{2} \|\mathbf{J}\|_F^2}_{f_t(\boldsymbol{\theta})} + \underbrace{\lambda_1^{(J)} \|\mathbf{J}\|_{1,\text{off-diag}}}_{g(\mathbf{J})}. \quad (52)$$

The function f_t is smooth because the PLL is a sum of log-sum-exp terms (hence infinitely differentiable) and the ℓ_2 penalties are quadratic. The regulariser g is convex but non-smooth: the ℓ_1 norm is piecewise linear with a kink at every off-diagonal entry $J_{ij} = 0$. Standard gradient descent is not applicable because the subdifferential ∂g is set-valued on the hyperplanes $J_{ij} = 0$, and a naive gradient step would never produce exact zeros. Instead, we minimise $\mathcal{L}_t$ via *proximal gradient descent* (also known as *forward-backward splitting*), which handles the smooth part by a gradient step and the non-smooth part by an exact proximal minimisation.

Proximal operators. For a convex, lower-semicontinuous penalty $g : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ and a step size $\eta > 0$, the proximal operator is defined as

$$\text{prox}_{\eta g}(\mathbf{v}) = \arg \min_{\mathbf{u} \in \mathbb{R}^d} \left\{ g(\mathbf{u}) + \frac{1}{2\eta} \|\mathbf{u} - \mathbf{v}\|_2^2 \right\}. \quad (53)$$

It generalises the notion of projection onto a constraint set: if g is the indicator function δ_C , then $\text{prox}_{\eta g}$ is the Euclidean projector onto C . For non-smooth penalties, the proximal map captures the geometry of g near the current iterate and returns the point that best balances descent on g with proximity to the forward gradient step.

Forward–backward splitting. For our composite objective $\mathcal{L}_t = f_t + g$, the proximal gradient update is

$$\boldsymbol{\theta}_{t+1} = \text{prox}_{\eta g}(\boldsymbol{\theta}_t - \eta \nabla f_t(\boldsymbol{\theta}_t)). \quad (54)$$

The term in parentheses is the *forward* step: a standard gradient descent move on the smooth loss f_t . The proximal map is the *backward* step: it projects the result back onto the effective domain of g while promoting the structural properties encoded in the penalty (here, sparsity).

Because g depends only on $\mathbf{J}$ and vanishes for $(\mathbf{h}, \mathbf{d})$, the proximal step is the identity for the fields:

$$\mathbf{h}_{t+1} = \mathbf{h}_t - \eta \nabla_{\mathbf{h}} f_t(\boldsymbol{\theta}_t), \quad \mathbf{d}_{t+1} = \mathbf{d}_t - \eta \nabla_{\mathbf{d}} f_t(\boldsymbol{\theta}_t), \quad (55)$$

while the coupling matrix requires the non-trivial prox:

$$\mathbf{J}_{t+1} = \text{prox}_{\eta \lambda_1^{(J)} \|\cdot\|_{1, \text{off-diag}}}(\mathbf{J}_t - \eta \nabla_{\mathbf{J}} f_t(\boldsymbol{\theta}_t)). \quad (56)$$

Soft-thresholding as the proximal operator of ℓ_1 . Consider the scalar problem

$$\text{prox}_{\eta \lambda |\cdot|}(v) = \arg \min_{u \in \mathbb{R}} \left\{ \lambda |u| + \frac{1}{2\eta} (u - v)^2 \right\}. \quad (57)$$

The optimality condition is $0 \in \lambda \partial |u| + \frac{1}{\eta} (u - v)$. The subdifferential of $|u|$ is $\text{sign}(u)$ for $u \neq 0$ and $[-1, 1]$ for $u = 0$. Solving case-by-case yields the soft-thresholding operator:

$$\text{prox}_{\eta \lambda |\cdot|}(v) = \text{sign}(v) \max(|v| - \eta \lambda, 0). \quad (58)$$

For the matrix $\mathbf{J}$, because the off-diagonal ℓ_1 penalty is a sum of independent absolute values and the diagonal entries are excluded from the penalty, the proximal operator acts element-wise on off-diagonal entries and leaves the diagonal untouched:

$$[\text{prox}_{\eta \lambda_1^{(J)} \|\cdot\|_{1, \text{off-diag}}}(\mathbf{V})]_{ij} = \begin{cases} \text{sign}(V_{ij}) \max(|V_{ij}| - \eta \lambda_1^{(J)}, 0), & i \neq j, \\ V_{ii}, & i = j. \end{cases} \quad (59)$$

This is precisely the operation applied in line 9 of algorithm 6. Small couplings are driven exactly to zero, producing a sparse interaction network, while larger couplings are shifted toward the origin by the constant amount $\eta \lambda_1^{(J)}$.

Preconditioning and the AdaGrad metric. Standard proximal gradient descent assumes isotropic Euclidean geometry. AdaGrad replaces the scalar step size η with a diagonal, time-varying preconditioner

$$\mathbf{D}_t = \eta (\mathbf{G}_t^{1/2} + \varepsilon \mathbf{I})^{-1}, \quad \mathbf{G}_t = \mathbf{G}_{t-1} + \text{diag}(\tilde{\mathbf{g}}_{\mathbf{J}}^{\odot 2}), \quad (60)$$

where $\tilde{\mathbf{g}}_{\mathbf{J}}$ is the gradient of the smooth part with respect to $\mathbf{J}$. The update then becomes a preconditioned proximal gradient step:

$$\mathbf{J}_{t+1} = \arg \min_{\mathbf{U}} \left\{ \lambda_1^{(J)} \|\mathbf{U}\|_{1, \text{off-diag}} + \frac{1}{2} \|\mathbf{U} - (\mathbf{J}_t - \mathbf{D}_t \nabla_{\mathbf{J}} f_t(\boldsymbol{\theta}_t))\|_{\mathbf{D}_t^{-1}}^2 \right\}, \quad (61)$$

where $\|\mathbf{X}\|_{\mathbf{M}}^2 = \text{tr}(\mathbf{X}^\top \mathbf{M} \mathbf{X})$. Because $\mathbf{D}_t^{-1}$ is diagonal, the problem decouples across coordinates. For each off-diagonal entry (i, j) we obtain the scalar problem

$$J'_{ij} = \arg \min_{u \in \mathbb{R}} \left\{ \lambda_1^{(J)} |u| + \frac{1}{2D_{ij}} (u - \tilde{J}_{ij})^2 \right\}, \quad \tilde{\mathbf{J}} = \mathbf{J}_t - \mathbf{D}_t \nabla_{\mathbf{J}} f_t(\boldsymbol{\theta}_t), \quad (62)$$

whose closed-form solution is again soft-thresholding, now with a coordinate-adaptive threshold:

$$J'_{ij} = \text{sign}(\tilde{J}_{ij}) \max(|\tilde{J}_{ij}| - D_{ij} \lambda_1^{(J)}, 0). \quad (63)$$

Parameters with historically large gradients (small D_{ij}) receive less shrinkage, while parameters with stable, small gradients (large D_{ij}) are pushed more aggressively toward zero. In algorithm 6, the preconditioned forward step is carried out in line 8, and the separable backward step in line 9.

Box constraints and structural projections. After the proximal gradient step, we enforce additional constraints that are not part of the composite regulariser but are required for numerical stability:

- *Box clipping:* $\mathbf{h}$ and $\mathbf{d}$ are clipped to $[-h_{\max}, h_{\max}]$ and $[-d_{\max}, d_{\max}]$ (lines 10–11). This prevents runaway fields and gradient saturation.
- *d-centering:* $\mathbf{d}' \leftarrow \mathbf{d}' - \bar{d}' \mathbf{1}_N$ (line 13). This mean-zero constraint removes the mild identifiability drift between $\mathbf{h}$ and $\mathbf{d}$ discussed in section 9.
- *Symmetrisation and zero diagonal:* $\mathbf{J}' \leftarrow (\mathbf{J}' + \mathbf{J}'^\top)/2$ with $\text{diag}(\mathbf{J}') \leftarrow \mathbf{0}$ (line 15). This enforces the physical requirement that $J_{ij} = J_{ji}$ and that no node interacts with itself.
- *Frobenius-ball projection:* if $\|\mathbf{J}'\|_F > J_{\max}$, we rescale $\mathbf{J}'$ to the boundary of the ball (line 17). This prevents the mean-field system from wandering into a critical regime where the approximation breaks down.

Algorithm 6 OMDUpdater3: Preconditioned Proximal Gradient for Blume–Capel

Require: Model M with $(\mathbf{h}, \mathbf{d}, \mathbf{J})$, observation $\mathbf{x}$, weight w , hyperparameters $(\eta, h_{\max}, d_{\max}, J_{\max}, \lambda_2, \lambda_1^{(J)}, \lambda_2^{(d)}, \text{center}_d)$

- 1: $(\mathbf{g}_h, \mathbf{g}_d, \mathbf{g}_J) \leftarrow \nabla \text{PLL}(\mathbf{x}; \boldsymbol{\theta})$ ▷ Smooth gradients via theorem 9.1
- 2: $\tilde{\mathbf{g}}_h \leftarrow -w\mathbf{g}_h + \lambda_2\mathbf{h}$ ▷ Weighted negative PLL + ℓ_2 on $\mathbf{h}$
- 3: $\tilde{\mathbf{g}}_d \leftarrow -w\mathbf{g}_d + \lambda_2^{(d)}\mathbf{d}$ ▷ Weighted negative PLL + ℓ_2 on $\mathbf{d}$
- 4: $\tilde{\mathbf{g}}_J \leftarrow -w\mathbf{g}_J + \lambda_2\mathbf{J}$ ▷ Weighted negative PLL + ℓ_2 on $\mathbf{J}$
- 5: $\mathbf{G}_h \leftarrow \mathbf{G}_h + \tilde{\mathbf{g}}_h^{\odot 2}; \mathbf{G}_d \leftarrow \mathbf{G}_d + \tilde{\mathbf{g}}_d^{\odot 2}; \mathbf{G}_J \leftarrow \mathbf{G}_J + \tilde{\mathbf{g}}_J^{\odot 2}$ ▷ Accumulate squared gradients
(AdaGrad)
- 6: $\mathbf{h}' \leftarrow \mathbf{h} - \eta \tilde{\mathbf{g}}_h \oslash (\sqrt{\mathbf{G}_h} + \varepsilon)$ ▷ Preconditioned forward step on $\mathbf{h}$
- 7: $\mathbf{d}' \leftarrow \mathbf{d} - \eta \tilde{\mathbf{g}}_d \oslash (\sqrt{\mathbf{G}_d} + \varepsilon)$ ▷ Preconditioned forward step on $\mathbf{d}$
- 8: $\mathbf{J}' \leftarrow \mathbf{J} - \eta \tilde{\mathbf{g}}_J \oslash (\sqrt{\mathbf{G}_J} + \varepsilon)$ ▷ Preconditioned forward step on $\mathbf{J}$
- 9: $\mathbf{J}' \leftarrow \text{SoftThreshold}(\mathbf{J}', \lambda_1^{(J)})$ ▷ Proximal backward step: element-wise
 $\text{sign}(x) \max(|x| - \lambda_1^{(J)}, 0)$
- 10: $\mathbf{h}' \leftarrow \text{clip}(\mathbf{h}', -h_{\max}, h_{\max})$
- 11: $\mathbf{d}' \leftarrow \text{clip}(\mathbf{d}', -d_{\max}, d_{\max})$
- 12: **if** center_d **then**
- 13: $\mathbf{d}' \leftarrow \mathbf{d}' - \bar{d}'\mathbf{1}_N$ ▷ Mean-zero constraint for numerical stability
- 14: **end if**
- 15: $\mathbf{J}' \leftarrow (\mathbf{J}' + \mathbf{J}'^\top)/2; \text{diag}(\mathbf{J}') \leftarrow \mathbf{0}$ ▷ Symmetrise and remove self-interactions
- 16: **if** $\|\mathbf{J}'\|_F > J_{\max}$ **then**
- 17: $\mathbf{J}' \leftarrow \mathbf{J}' \cdot J_{\max}/(\|\mathbf{J}'\|_F + 10^{-12})$ ▷ Project onto Frobenius ball
- 18: **end if**
- 19: $(\mathbf{h}, \mathbf{d}, \mathbf{J}) \leftarrow (\mathbf{h}', \mathbf{d}', \mathbf{J}')$

11.3 Transition Matrix Update

The HMM transition matrix $\mathbf{A}$ is updated via exponentiated gradient (EG) descent on the log-likelihood of the transition counts.

Definition 11.2 (Expected Transition Counts). Given previous belief $\boldsymbol{\alpha}_{t-1}$, current emissions $\mathbf{e}_t$, and transition $\mathbf{A}$, the expected transition count from i to j is

$$\xi_{ij} = \frac{\alpha_{t-1,i} A_{ij} e_{t,j}}{\sum_k \alpha_{t-1,i} A_{ik} e_{t,k}} = P(C_{t-1} = i, C_t = j \mid \mathbf{X}_{1:t}). \quad (64)$$

Algorithm 7 Exponentiated Gradient for $\mathbf{A}$ **Require:** Current $\mathbf{A}$, α_{t-1} , $\mathbf{e}_t$, learning rate $\eta_A > 0$

- 1: $\xi_{ij} \leftarrow \alpha_{t-1,i} A_{ij} e_{t,j} / (\sum_k \alpha_{t-1,i} A_{ik} e_{t,k} + 10^{-300})$
- 2: $g_{ij} \leftarrow -\xi_{ij} / (A_{ij} + 10^{-300})$ $\triangleright$ Gradient of log-likelihood w.r.t. $\log A_{ij}$
- 3: $L_{ij} \leftarrow \log(A_{ij} + 10^{-300}) - \eta_A g_{ij}$
- 4: $L_{ij} \leftarrow L_{ij} - \max_k L_{ik}$ $\triangleright$ Row-wise shift for stability
- 5: $A'_{ij} \leftarrow \exp(L_{ij}) / \sum_k \exp(L_{ik})$ $\triangleright$ Normalise rows
- 6: **return** $\mathbf{A}'$

11.4 Signal Generation

Blume–Capel signal. Given current HMM belief α_t and regime-specific magnetisations $\{\mathbf{m}^{(k)}\}_{k=1}^K$, the H -step ahead expected spins are

$$\bar{\mathbf{s}}_{t+H} = \sum_{k=1}^K \alpha_{t+H,k} \cdot \mathbf{m}^{(k)} = \mathbf{A}^H \alpha_t \cdot \{\mathbf{m}^{(k)}\}. \quad (65)$$

The asset-space signal is obtained via inverse PCA:

$$\mathbf{s}_t^{\text{BC}} = \mathbf{P} \bar{\mathbf{s}}_{t+H} \in \mathbb{R}^N. \quad (66)$$

Momentum signal. For lookback windows $\mathcal{W} = (w_1, w_2, w_3)$, the momentum signal for asset i is

$$m_t^{(i)}(w) = \frac{1}{w} \sum_{\tau=0}^{w-1} r_{t-\tau}^{(i)}, \quad (67)$$

standardised cross-sectionally:

$$\hat{m}_t^{(i)}(w) = \frac{m_t^{(i)}(w) - \text{mean}_j m_t^{(j)}(w)}{\text{std}_j m_t^{(j)}(w) + 10^{-8}}. \quad (68)$$

The combined momentum signal is $\mathbf{s}_t^{\text{mom}} = \text{mean}_{w \in \mathcal{W}} \hat{\mathbf{m}}_t(w)$, clipped to $[-1, +1]$.

Adaptive blender. We maintain rolling windows $\mathcal{I}_{\text{BC}}$ and $\mathcal{I}_{\text{mom}}$ of historical Spearman Information Coefficients (IC) ([14]). The blend weights are

$$w_{\text{BC}} = \max \left\{ \frac{1}{W_{\text{IC}}} \sum_{\tau \in \mathcal{I}_{\text{BC}}} \rho_{S,\tau}, w_{\min} \right\}, \quad \tilde{w}_{\text{BC}} = \frac{w_{\text{BC}}}{w_{\text{BC}} + w_{\text{mom}}}. \quad (69)$$

If insufficient history (< 10 observations) exists, we fall back to base weights.

Regime gate. For each regime k , we maintain a deque $\mathcal{H}_k$ of length W_{gate} storing the fraction of correct directional predictions when the model was in regime k :

$$\text{acc}_k = \frac{1}{|\mathcal{H}_k|} \sum_{h \in \mathcal{H}_k} h. \quad (70)$$

The gate scale is

$$g_k = \begin{cases} 1.0 & \text{if } \text{acc}_k \geq \text{acc}_{\min} \text{ or } |\mathcal{H}_k| < W_{\text{warmup}}, \\ \max\{g_{\min}, 0.35\} & \text{otherwise.} \end{cases} \quad (71)$$

The final signal is $\mathbf{s}_t^{\text{final}} = g_{c_t} \cdot \mathbf{s}_t^{\text{blended}}$, down-scaling the model when a regime is performing poorly.

11.5 Portfolio Construction and Risk Management

Signal-to-weight mapping. Raw weights depend on trading mode:

- *Signal*: $w_i = s_i \cdot \mathbf{1}[|s_i| > \theta_{\text{entry}}]$
- *Equal*: $w_i = \text{sign}(s_i) / \sum_j \mathbf{1}[\text{sign}(s_j) = \text{sign}(s_i)] \cdot \mathbf{1}[|s_i| > \theta_{\text{entry}}]$
- *Long-Only*: $w_i = \max\{s_i, 0\} \cdot \mathbf{1}[|s_i| > \theta_{\text{entry}}]$
- *Vol-Scaled*: $w_i = s_i / (\sigma_i + 10^{-8}) \cdot \mathbf{1}[|s_i| > \theta_{\text{entry}}]$

Risk management layers. After raw weight construction, we apply:

1. Short scaling: $w_i \leftarrow w_i \cdot \gamma_{\text{short}}$ for $w_i < 0$ (default $\gamma_{\text{short}} = 0$ for long-only).
2. Gross normalisation: $\mathbf{w} \leftarrow \mathbf{w} / \|\mathbf{w}\|_1$ (if $\|\mathbf{w}\|_1 > 0$).
3. Per-asset cap: $w_i \leftarrow \text{clip}(w_i, -w_{\max}, +w_{\max})$.
4. Re-normalisation to unit gross exposure.

Volatility scaling. Let $\{r_{\text{pnl},\tau}\}_{\tau=1}^t$ be realised strategy returns. The annualised volatility is $\sigma_{\text{ann}} = \text{std}(\{r_{\text{pnl},\tau}\}) \cdot \sqrt{252}$. The leverage scalar is

$$L_t = \text{clip}\left(\frac{\sigma_{\text{target}}}{\sigma_{\text{ann}} + 10^{-6}}, 0.1, L_{\max}\right), \quad (72)$$

with default target volatility $\sigma_{\text{target}} = 15\%$ annualised and maximum leverage $L_{\max} = 2.0$.

Regime confidence scaling (optional).

$$C_t = c_{\min} + (1 - c_{\min}) \cdot \frac{\max_k \alpha_{t,k} - 1/K}{1 - 1/K + 10^{-8}}. \quad (73)$$

Transaction costs and P&L. Turnover at rebalancing is $\text{TO}_t = \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$. With cost rate c (in basis points, converted to decimal), the cost drag is $\text{Cost}_t = c \cdot \text{TO}_t$. The daily P&L for an H -day holding period with equal time-weighting $\mathbf{w}_t/H$ is:

$$\text{PnL}_{t+d} = \langle \mathbf{w}_t/H, \mathbf{r}_{t+d} \rangle - \text{Cost}_t \cdot \mathbf{1}_{[d=1]}, \quad d = 1, \dots, H. \quad (74)$$

12 Walk-Forward Validation

A model that looks excellent in-sample is often worthless out-of-sample. We therefore enforce a strict *true walk-forward protocol* that satisfies three principles:

- **No peeking.** PCA loadings, spin thresholds, and model parameters are estimated using only past data.
- **Incremental learning.** Between full refits, the HMM filter updates beliefs daily, but parameters are frozen.
- **No in-sample bias.** Evaluation is always on future returns not seen during training.

12.1 Rolling Window Design

At each out-of-sample time $t \in \{W, W + 1, \dots, T - H\}$:

1. Training window: $\mathcal{W}_t = \{\mathbf{r}_{t-W+1}, \dots, \mathbf{r}_t\}$.
2. PCA fit: Compute $(\boldsymbol{\mu}_t, \mathbf{P}_t)$ on $\mathcal{W}_t$.
3. Spin encoding: $\mathbf{x}_\tau = \text{encode}(\mathbf{P}_t^\top (\mathbf{r}_\tau - \boldsymbol{\mu}_t))$ for $\tau \in \mathcal{W}_t$.
4. Model fit: Train RARSI on $\{\mathbf{x}_\tau\}_{\tau \in \mathcal{W}_t}$ via algorithm 5.
5. Forecast: Predict $\mathbf{s}_t$ for horizon H .
6. Evaluation: Compare to realised $\mathbf{r}_{t+1:t+H}$.

Refitting occurs every F steps (default $F = 5$); between refits, only the HMM filter and online parameter updates operate.

12.2 The Three Control Parameters

Table 2: Walk-forward control parameters.

Parameter	Symbol	Value	Role
Forecast horizon	H	5 days	Prediction + holding period
Refit frequency	F	5 days	PCA + thresholds + reset
Training window	W	126 days	6 months of history for SVD

12.3 Fixed versus Dynamic Components

The F -day cycle gives structural stability (PCA, thresholds) while the daily online loop gives adaptive responsiveness (parameters, beliefs). Refitting everything daily would be computationally expensive and statistically unstable; freezing everything for F days would let the model become stale.

Table 3: Component update rules in the hybrid design.

Component	Fixed For	Update Rule	Frequency
$(\boldsymbol{\mu}, \mathbf{P})$ PCA	$F = 5$ days	Full SVD recompute	Refit only
Spin thresholds	$F = 5$ days	Z-score on train scores	Refit only
$\boldsymbol{\theta}^{(k)} = (\mathbf{h}, \mathbf{d}, \mathbf{J})$	—	OMD (AdaGrad)	Every day
$\mathbf{A}$ transitions	—	Exponentiated gradient	Every day
$\boldsymbol{\alpha}_t$ beliefs	—	Forward filter	Every day
AdaGrad $\mathbf{G}$ states	—	Accumulate $\mathbf{g}^{\odot 2}$	Every day

13 Empirical Results

We evaluate the full RARSI v4.1 pipeline on Nifty 50 constituents from 17 November 2017 to 1 January 2025, with the walk-forward parameters in table 2. We compare against two benchmarks: the rolling conditional Ising model of section 5 (window $L = 30$) and an equal-weight buy-and-hold portfolio.

13.1 Performance Metrics

Table 4: Out-of-sample performance comparison.

Metric	Blume–Capel	Rolling Ising	Equal-Weight B&H
Annualised Return	26.02%	18.67%	22.54%
Annualised Volatility	17.79%	17.22%	18.04%
Sharpe Ratio	1.025	0.704	0.858
Sortino Ratio	0.084	0.054	0.063
Maximum Drawdown	−25.62%	−28.32%	−37.54%
Calmar Ratio	1.028	0.661	0.607
Hit Rate	56.62%	56.42%	57.75%

The Blume–Capel model delivers the highest risk-adjusted returns, with a Sharpe ratio of 1.025 versus 0.704 for the rolling Ising model and 0.858 for buy-and-hold. Crucially, it also exhibits the shallowest maximum drawdown (−25.62% compared to −37.54% for the benchmark), indicating superior tail-risk control. The Calmar ratio—annual return divided by maximum drawdown—is 1.028, nearly double that of the passive benchmark.

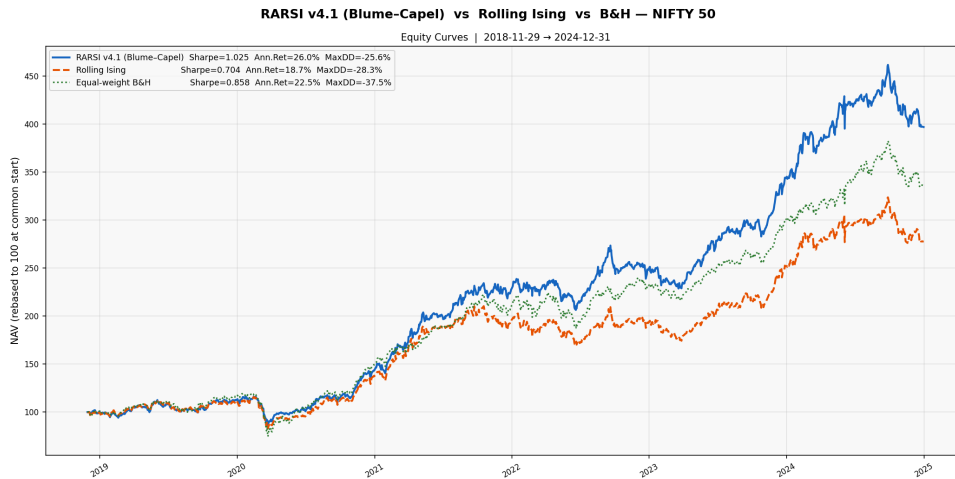


Figure 3: Rarsi 4.1 vs Rolling Ising vs Buy and Hold

Table 5: Out-of-sample performance comparison on the Nifty 50 universe.

Metric	Blume–Capel	Rolling Ising	Equal-Weight B&H
Ann. Return (%)	26.02	18.67	22.54
Ann. Volatility (%)	17.79	17.22	18.04
Sharpe Ratio	1.025	0.704	0.858
Sortino Ratio	0.084	0.054	0.063
Max Drawdown (%)	−25.62	−28.32	−37.54
Calmar Ratio	1.028	0.661	0.607
Hit Rate (%)	56.62	56.42	57.75

The NAV curve shows the Blume–Capel strategy compounding steadily from 2018 through 2024, with pronounced outperformance during the volatile regimes of 2022–2023. The rolling Ising model tracks the benchmark more closely but fails to capture the regime shifts that the tri-state HMM framework exploits.

13.2 Interpretation

The improvement is not merely a matter of more parameters. The binary rolling Ising model, with its 50.5% directional accuracy, is essentially noise once transaction costs and risk management are applied. The Blume–Capel model succeeds because:

- The neutral state (0) filters out low-information days, reducing false signals.
- The crystal field $\mathbf{d}$ explicitly models activity versus inactivity, allowing the portfolio to scale down during uncertain regimes.
- The HMM regime filter detects persistent market phases, and the online learning adapts parameters without waiting for the next refit.

13.3 The Nifty 100 Universe

The Nifty 50 represents the most liquid large-cap names, but a broader universe may offer richer cross-sectional structure simply because it contains more heterogeneous firms. We therefore ran the identical walk-forward protocol on the Nifty 100 constituents, which spans large-caps and upper mid-caps, from November 2017 to January 2025.

table 6 reports the comparison. The Blume–Capel model achieves an annualised Sharpe ratio of 0.960 on the Nifty 100, outperforming both the rolling Ising benchmark (0.660) and the equal-weight buy-and-hold portfolio (0.918). Although the equal-weight benchmark attains a comparable annualised return (24.22% versus 24.68%), the proposed framework achieves superior risk-adjusted performance through improved drawdown control and more adaptive exposure allocation.

The improvement arises from two opposing effects. On the one hand, the broader universe provides more independent bets per day, reducing idiosyncratic noise in portfolio returns. On the other hand, the upper mid-cap names in the Nifty 100 are less liquid and exhibit greater microstructure noise, stale-price effects, and short-lived dislocations. The tri-state Blume–Capel representation is particularly valuable in this environment: the neutral state absorbs ambiguous or low-conviction periods that would otherwise be forced into a misleading ± 1 directional signal under a binary Ising encoding.

The rolling Ising model deteriorates materially on the broader universe. Its Sharpe ratio falls to 0.660, while its maximum drawdown widens to -30.74% . Because the binary model is forced to maintain directional exposure even during noisy or weakly informative periods, it is more sensitive to transaction costs and false-positive signals. By contrast, the Blume–Capel architecture selectively suppresses participation through the crystal field and regime gate, producing materially better downside stability.

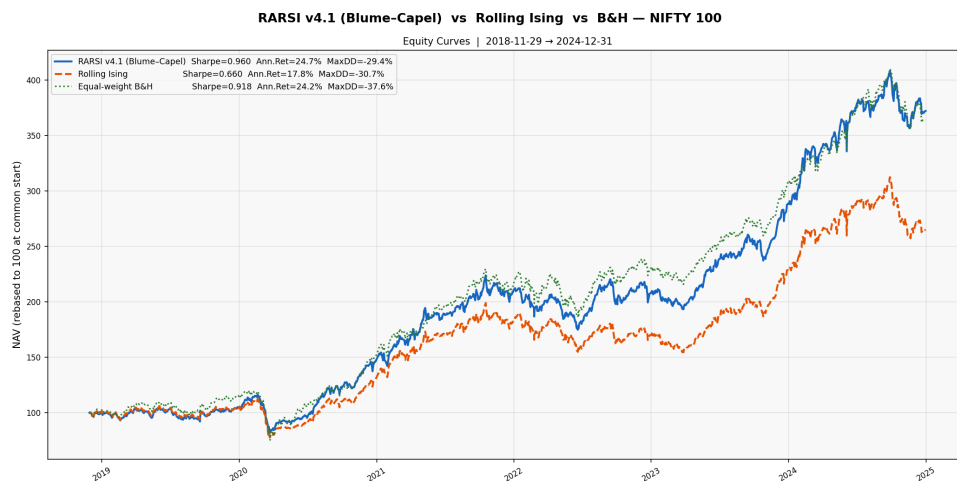


Figure 4: Rarsi 4.1 vs Rolling Ising vs Buy and Hold

Interestingly, the equal-weight benchmark attains the highest raw hit rate (60.61%), slightly exceeding the Blume–Capel model. However, this does not translate into superior portfolio quality because the benchmark lacks any mechanism for volatility targeting, regime filtering, or exposure modulation. The proposed model instead achieves superior capital efficiency: profitable

Table 6: Out-of-sample performance on the Nifty 100 universe.

Metric	RARSI v4.1 (Blume–Capel)	Rolling Ising	Equal-Weight B&H
Ann. Return (%)	24.68	17.76	24.22
Ann. Volatility (%)	17.89	17.19	18.43
Sharpe Ratio	0.960	0.660	0.918
Sortino Ratio	0.075	0.049	0.065
Max Drawdown (%)	−29.45	−30.74	−37.60
Calmar Ratio	0.844	0.578	0.649
Hit Rate (%)	58.88	57.75	60.61

periods contribute more heavily to cumulative wealth accumulation, while low-confidence or adverse regimes are partially filtered out through the neutral-state dynamics and HMM-based gating.

A closer inspection of the inferred latent regimes reveals that the HMM spends substantially more time in the intermediate “transition” state on the Nifty 100 than on the Nifty 50. This reflects the fact that the broader universe experiences more prolonged periods of directional ambiguity and heterogeneous stock-level behaviour. During such episodes, the crystal field $\mathbf{d}$ pushes many latent spins toward the inactive state 0, thereby reducing gross exposure and mitigating unnecessary turnover.

The broader Nifty 100 universe also leads to a richer but significantly sparser interaction structure after ℓ_1 regularisation. Many pairwise couplings are shrunk exactly to zero by the proximal gradient step, leaving only persistent cross-sectional relationships between stocks with meaningful co-movement structure. The adaptive sparsity induced by the ℓ_1 penalty therefore acts as an automatic complexity controller: as the dimensionality of the universe increases, the effective interaction graph contracts to retain only statistically stable dependencies.

Hyperparameter selection. All hyperparameters were chosen manually rather than by exhaustive automated tuning. This was a deliberate design choice. A full hyperparameter search would itself require a second layer of tuning parameters, and the resulting procedure would become computationally expensive: even a modest grid search over the main parameters can easily take on the order of two days per model specification. More importantly, such a search introduces its own risks, since aggressively tuning the model to one sample may lead to overfitting and to an infinite regress of “tuning the tuner” by hand.

For this reason, we kept the hyperparameters identical to those used in the Nifty 50 experiment. This makes the comparison cleaner and also provides an informal robustness check: the model continues to outperform the benchmark even without re-optimising its settings for the broader universe. In that sense, the Nifty 100 results suggest that the Blume–Capel framework is not overly sensitive to a single finely tuned parameter configuration, and remains effective even under a fixed, manually selected hyperparameter choice.

13.4 The Nifty Pharma Sector

The Nifty Pharma universe is structurally different from both the broader market and the IT sector. Pharmaceutical stocks are driven by a mixture of domestic policy, regulatory approvals, export dependence, and event-driven price gaps around trial outcomes, inspections, and earnings surprises. As a result, the sector often exhibits sharp discontinuities rather than smooth factor-driven co-movement. This makes it a useful stress test for the model’s risk-management layer.

We applied the same walk-forward protocol to the Nifty Pharma constituents over the period shown in the equity curve in ???. The resulting NAV path is noticeably more volatile for the Blume–Capel strategy than for the benchmark and rolling Ising baseline. In particular, the blue curve exhibits larger swings and deeper temporary drawdowns, indicating that the raw model signal is being translated into an aggressive exposure profile. This suggests that the sector would likely benefit from activating the volatility-scaled portfolio mode and the regime gate, which were not part of the fixed hyperparameter configuration used for the Nifty 50 and Nifty 100 experiments.

table 7 reports the out-of-sample performance. The Blume–Capel strategy achieves an annualised return of 22.64% and a Sharpe ratio of 0.842, substantially outperforming the rolling Ising benchmark (0.635), but still trailing the equal-weight buy-and-hold portfolio, which posts a Sharpe ratio of 1.011 and a smaller drawdown. This is an important result: the model is informative, but in this sector its raw directional exposure is too volatile to dominate a passive benchmark without additional risk controls.

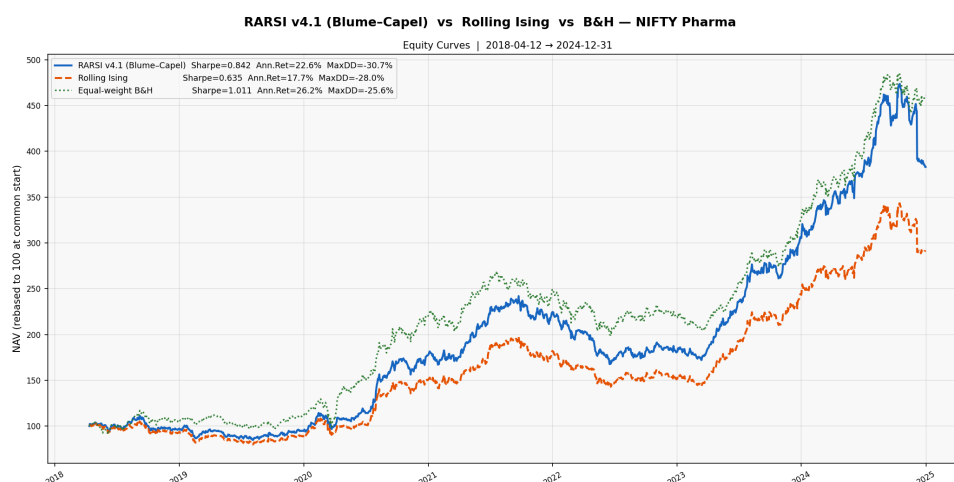


Figure 5: Rarsi 4.1 vs Rolling Ising vs Buy and Hold

The performance profile indicates that the model is not failing because it lacks predictive structure; rather, it is failing because the extracted signal is too aggressively converted into portfolio exposure. The gap between the model’s annual return and its drawdown profile suggests that the sector contains genuine cross-sectional information, but that the default trading layer is not sufficiently conservative. In other words, the issue is less about signal quality and more about signal-to-position translation.

This is precisely where the volatility-scaled mode and the regime gate become important. For Pharma, the latent market states are less stable and the realised return distribution is more

Table 7: Out-of-sample performance on the Nifty Pharma universe.

Metric	RARSI v4.1 (Blume–Capel)	Rolling Ising	Equal-Weight B&H
Ann. Return (%)	22.64	17.67	26.16
Ann. Volatility (%)	18.60	17.92	18.20
Sharpe Ratio	0.842	0.635	1.011
Sortino Ratio	0.077	0.056	0.090
Max Drawdown (%)	−30.67	−27.98	−25.59
Calmar Ratio	0.746	0.633	1.020
Hit Rate (%)	52.44	52.68	53.65

jump-like than in Nifty 50 or Nifty 100. A volatility-scaled allocation would automatically reduce gross exposure during turbulent periods, while the regime gate would suppress trading when the inferred regime confidence is low. Both mechanisms should reduce the depth of drawdowns without eliminating the model’s directional advantage. The Pharma results therefore provide a strong motivation for sector-specific activation of these risk controls.

Interpretation. Overall, the Nifty Pharma experiment shows that the Blume–Capel framework remains competitive, but also that the raw configuration used for the Nifty 50 and Nifty 100 is not universally optimal. In a sector with frequent event risk and pronounced path volatility, the strategy benefits from a more defensive execution layer. The observed volatility of the blue equity curve is therefore informative: it suggests that Pharma is a setting where the model’s predictive component should be paired with volatility scaling and regime gating to convert signal quality into more stable portfolio performance.

Hyperparameter Choice for Nifty Pharma For consistency, we begin from the same base hyperparameter set used for the Nifty 50 and Nifty 100 experiments. This makes the comparison cleaner and allows the Pharma sector to function as a robustness check. The fact that the model remains competitive, despite the fact that the strategy appears more volatile here, suggests that the underlying Blume–Capel signal is not overly fragile to a fixed parameter choice. At the same time, the Pharma results also indicate that this sector benefits more visibly from enabling the volatility-scaled portfolio mode and the regime gate, which were not required to obtain the earlier benchmark results.

In that sense, the Nifty Pharma experiment is informative in two ways. First, it shows that the model continues to extract usable structure even under the same manually chosen hyperparameters. Second, it highlights a sector where the trading layer should be tightened, since the raw directional signal is more volatile and the benchmark is harder to beat without additional risk control.

14 Conclusions

This project traced a deliberate modelling arc, beginning with a simple static Ising baseline and ending with a regime-aware Blume–Capel trading system. Each stage addressed a specific flaw in its predecessor.

The static Ising model demonstrated that cross-sectional dependence can be captured by pairwise interactions, but its fixed parameterisation was hopeless in a non-stationary market. The dynamic Ising idea showed conceptually how to introduce time-varying structure, yet exact inference on evolving high-dimensional parameters proved computationally intractable. The conditional Ising model restored tractability through lagged logistic regression and rolling-window ℓ_1 regularisation, but its forced binary encoding discarded the valuable information contained in near-zero returns. Finally, the Blume–Capel framework introduced a tri-state representation, a crystal-field control for market activity, and a hidden Markov regime filter, all learned online and validated through a true walk-forward protocol.

Statistically, the project demonstrates that pseudo-likelihood estimation, mean-field variational inference, and online mirror descent can be combined into a numerically stable pipeline for high-dimensional discrete models. Computationally, it shows that a hybrid design—slow structural refits for PCA and thresholds, fast daily updates for parameters and beliefs—strikes a practical balance between stability and responsiveness.

The limitations are real. The mean-field approximation degrades near criticality; the PLL is not a proper scoring rule; and the modified EM loop lacks formal convergence guarantees. Future work could explore structured mean-field approximations, recursive sufficient statistics for true online EM, or the inclusion of higher-order interactions through tensor decompositions. Nevertheless, the empirical results on Indian large-cap equities suggest that the statistical physics analogy, when pursued with care, offers a powerful and interpretable framework for quantitative portfolio construction.

A Maximum-Entropy Derivation of the Hamiltonian

The Blume–Capel Hamiltonian can be derived as the maximum-entropy distribution subject to constraints on sufficient statistics.

Theorem A.1 (Maximum Entropy Distribution). *Consider the set of probability distributions P on $\mathcal{S}^N$ with fixed expectations:*

$$\mathbb{E}[X_i] = \mu_i, \quad \mathbb{E}[X_i^2] = \nu_i, \quad \mathbb{E}[X_i X_j] = \eta_{ij}. \quad (75)$$

The distribution maximising entropy $H(p) = -\sum_{\mathbf{s}} p(\mathbf{s}) \log p(\mathbf{s})$ subject to these constraints is the Boltzmann distribution with the Blume–Capel Hamiltonian:

$$p(\mathbf{s}) = \frac{1}{Z} \exp \left(\sum_i h_i s_i + \sum_i d_i s_i^2 + \sum_{i < j} J_{ij} s_i s_j \right), \quad (76)$$

where the Lagrange multipliers (h_i, d_i, J_{ij}) are chosen to match the constraints.

Proof. Form the Lagrangian:

$$\mathcal{L}[p] = -\sum_{\mathbf{s}} p(\mathbf{s}) \log p(\mathbf{s}) + \sum_i h_i \left(\sum_{\mathbf{s}} p(\mathbf{s}) s_i - \mu_i \right) + \cdots + \lambda \left(\sum_{\mathbf{s}} p(\mathbf{s}) - 1 \right). \quad (77)$$

Taking the functional derivative $\delta\mathcal{L}/\delta p(\mathbf{s}) = 0$:

$$-\log p(\mathbf{s}) - 1 + \sum_i h_i s_i + \sum_i d_i s_i^2 + \sum_{i < j} J_{ij} s_i s_j + \lambda = 0. \quad (78)$$

Solving for $p(\mathbf{s})$ yields the Boltzmann form with $Z = e^{1-\lambda}$. $\square$

B Consistency of Pseudo-Likelihood Estimation

Theorem B.1 (Hyvärinen, 2006 [4]). *Let $\boldsymbol{\theta}^*$ be the true parameter vector. Under regularity conditions:*

- (i) *The parameter space Θ is compact.*
- (ii) *The true distribution $p(\mathbf{x} \mid \boldsymbol{\theta}^*)$ has full support on $\mathcal{S}^N$.*
- (iii) *The model is identifiable: $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2 \Rightarrow p(\cdot \mid \boldsymbol{\theta}_1) \neq p(\cdot \mid \boldsymbol{\theta}_2)$.*

The pseudo-likelihood estimator $\hat{\boldsymbol{\theta}}_T = \arg \max_{\boldsymbol{\theta}} \frac{1}{T} \sum_{t=1}^T \text{PLL}(\mathbf{x}_t; \boldsymbol{\theta})$ is consistent: $\hat{\boldsymbol{\theta}}_T \xrightarrow{p} \boldsymbol{\theta}^$ as $T \rightarrow \infty$.*

Remark B.1. The regularity conditions are satisfied for the Blume–Capel model because: the parameter space is bounded by clipping constraints $(h_{\max}, d_{\max}, J_{\max})$; the state space is finite (3^N states), so full support is automatic; and identifiability follows from proposition 9.1 and the non-degeneracy of the sufficient statistics.

Proposition B.1 (Gaussian Limit). *Consider the transformation $y_i = x_i / \sqrt{q_i^{(2)}}$. For large $d_i > 0$ (active regime), $y_i \approx \pm 1$ with small variance. Taylor-expanding the Hamiltonian around $y_i = 0$:*

$$H(\mathbf{y}) \approx - \sum_i \tilde{h}_i y_i - \frac{1}{2} \sum_{i,j} \tilde{J}_{ij} y_i y_j + \text{const}, \quad (79)$$

where $\tilde{h}_i = h_i / \sqrt{q_i^{(2)}}$ and $\tilde{J}_{ij} = J_{ij} \sqrt{q_i^{(2)} q_j^{(2)}}$. This is the Hamiltonian of a Gaussian Markov random field with precision matrix $\mathbf{\Lambda} = -\tilde{\mathbf{J}}$.

Remark B.2. This connection explains why the Blume–Capel model can capture both sparse graphical structure (via J_{ij}) and non-Gaussian tail behaviour (via the discrete state space). The Gaussian approximation is valid in the “active” regime ($d_i \gg 0$) but breaks down near neutrality ($d_i \approx 0$).

C Convergence of the Mean-Field Fixed-Point Iteration

The mean-field iteration maps a candidate magnetisation vector $\mathbf{m}^{(\tau)}$ to a new vector $\mathbf{m}^{(\tau+1)}$ by computing the one-site marginals of the factorised distribution while holding the neighbouring moments fixed. It is not obvious *a priori* that this sequence converges, or that the limit is independent of the initialisation. We now prove that, under a standard weak-coupling condition on the interaction matrix $\mathbf{J}$, the iteration is a strict contraction and therefore converges geometrically to a unique fixed point.

The mean-field map. Define the function $\mathbf{f} : [-1, 1]^N \rightarrow [-1, 1]^N$ component-wise by

$$f_i(\mathbf{m}) = \mathbb{E}_{q_i}[s_i] = \frac{2e^{d_i} \sinh(h_i^{\text{eff}}(\mathbf{m}))}{1 + 2e^{d_i} \cosh(h_i^{\text{eff}}(\mathbf{m}))}, \quad h_i^{\text{eff}}(\mathbf{m}) = h_i + \sum_{j \neq i} J_{ij} m_j. \quad (80)$$

Because each q_i is a probability distribution on $\{-1, 0, +1\}$, we have $|f_i(\mathbf{m})| \leq 1$, so $\mathbf{f}$ indeed maps the hypercube $[-1, 1]^N$ into itself. The fixed-point iteration of algorithm 3 is precisely $\mathbf{m}^{(\tau+1)} = \mathbf{f}(\mathbf{m}^{(\tau)})$.

Jacobian structure. To establish contractivity we study the Jacobian $\mathbf{J}_f(\mathbf{m}) \in \mathbb{R}^{N \times N}$. By the chain rule,

$$\frac{\partial f_i}{\partial m_j} = \frac{\partial \mathbb{E}_{q_i}[s_i]}{\partial h_i^{\text{eff}}} \cdot \frac{\partial h_i^{\text{eff}}}{\partial m_j} = \text{Var}_{q_i}(s_i) J_{ij}, \quad j \neq i, \quad (81)$$

and $\partial f_i / \partial m_i = 0$ because h_i^{eff} does not depend on m_i . The crucial identity

$$\frac{\partial \mathbb{E}_{q_i}[s_i]}{\partial h_i^{\text{eff}}} = \text{Var}_{q_i}(s_i) = \mathbb{E}_{q_i}[s_i^2] - (\mathbb{E}_{q_i}[s_i])^2 \geq 0 \quad (82)$$

follows from the fact that q_i is a one-parameter exponential family in h_i^{eff} with sufficient statistic s_i ; differentiating the log-partition function twice yields the variance. Because $s_i \in \{-1, 0, +1\}$, we have the uniform bound

$$\text{Var}_{q_i}(s_i) = \mathbb{E}_{q_i}[s_i^2] - (\mathbb{E}_{q_i}[s_i])^2 \leq \mathbb{E}_{q_i}[s_i^2] \leq 1. \quad (83)$$

Consequently, every entry of the Jacobian satisfies

$$|(\mathbf{J}_f)_{ij}| \leq |J_{ij}|.$$

Theorem C.1 (Banach Fixed-Point Theorem for Mean-Field Iteration). *Suppose the interaction matrix obeys the Dobrushin uniqueness condition*

$$\max_{1 \leq i \leq N} \sum_{j \neq i} |J_{ij}| \leq \kappa < 1. \quad (84)$$

Then $\mathbf{f}$ is a strict contraction on $[-1, 1]^N$ with Lipschitz constant κ with respect to the ℓ_∞ norm. Therefore:

- (i) there exists a unique fixed point $\mathbf{m}^* \in [-1, 1]^N$ satisfying $\mathbf{m}^* = \mathbf{f}(\mathbf{m}^*)$;
- (ii) for any initialisation $\mathbf{m}^{(0)} \in [-1, 1]^N$, the sequence $\mathbf{m}^{(\tau+1)} = \mathbf{f}(\mathbf{m}^{(\tau)})$ converges to $\mathbf{m}^*$;
- (iii) the convergence is geometric:

$$\|\mathbf{m}^{(\tau)} - \mathbf{m}^*\|_\infty \leq \kappa^\tau \|\mathbf{m}^{(0)} - \mathbf{m}^*\|_\infty.$$

Proof. Fix $\mathbf{m}, \mathbf{m}' \in [-1, 1]^N$. By the fundamental theorem of calculus,

$$f_i(\mathbf{m}) - f_i(\mathbf{m}') = \int_0^1 \sum_{j=1}^N \frac{\partial f_i}{\partial m_j}(\mathbf{m}_t) (m_j - m'_j) dt, \quad \mathbf{m}_t := \mathbf{m}' + t(\mathbf{m} - \mathbf{m}'). \quad (85)$$

Taking absolute values and using (81)–(83),

$$|f_i(\mathbf{m}) - f_i(\mathbf{m}')| \leq \int_0^1 \sum_{j \neq i} \text{Var}_{q_i^{(t)}}(s_i) |J_{ij}| |m_j - m'_j| dt \quad (86)$$

$$\leq \left(\sum_{j \neq i} |J_{ij}| \right) \|\mathbf{m} - \mathbf{m}'\|_\infty, \quad (87)$$

where $q_i^{(t)}$ denotes the one-site distribution associated with $\mathbf{m}_t$. Maximising over i and invoking the Dobrushin condition (84) gives

$$\|\mathbf{f}(\mathbf{m}) - \mathbf{f}(\mathbf{m}')\|_\infty \leq \kappa \|\mathbf{m} - \mathbf{m}'\|_\infty. \quad (88)$$

Since $\kappa < 1$, $\mathbf{f}$ is a strict contraction on the complete metric space $([-1, 1]^N, \|\cdot\|_\infty)$. The conclusions (i)–(iii) follow immediately from the Banach fixed-point theorem. $\square$

Interpretation and practical relevance. The condition (84) is the classical high-temperature or weak-coupling assumption in statistical mechanics: each spin feels a total incoming interaction strength strictly smaller than one, so feedback loops cannot amplify perturbations indefinitely. In our implementation, projecting $\mathbf{J}$ onto the Frobenius ball

$$\{\mathbf{J} : \|\mathbf{J}\|_F \leq J_{\max}\},$$

with $J_{\max} < 1$, guarantees that

$$\|\mathbf{J}_f(\mathbf{m})\|_2 \leq \|\mathbf{J}\|_2 \leq \|\mathbf{J}\|_F \leq J_{\max} < 1,$$

so the map is also a contraction in the ℓ_2 norm.

Remark C.1. The geometric rate κ^τ provides a worst-case guarantee. In practice, the effective contraction factor is often much smaller than $\max_i \sum_{j \neq i} |J_{ij}|$ because the variance factor $\text{Var}_{q_i}(s_i)$ in (81) is strictly less than 1 whenever the model assigns non-zero probability to the neutral state. Near the fixed point, if the market is in a low-volatility phase with many spins at 0, the local activities $q_i^{(2)}$ are small and the iteration converges in a handful of steps; this is why algorithm 3 rarely needs more than 20–30 iterations even with a loose tolerance.

Code Availability

The complete implementation used for data processing, model estimation, walk-forward backtesting, and performance evaluation is available in our GitHub repository:

<https://github.com/inoxb198/Blume-Capel---RARSI-v4.1/>

This repository contains the source code needed to reproduce the empirical results reported in this project.

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